

Day 3

DNS Server Configuration Theory (30 min) : How DNS resolves domain names to Ips Practical (30 min) : Configure a DNS server and test domain resolution Task: Document the configuration and troubleshooting steps.

1. Install and set up a DNS server on your local machine or a VM using BIND (Linux) or Simple DNS Plus (Windows), then create a new DNS zone for the domain musicmania.in.

ANS :

- **Objective**

Install a DNS server and create a DNS zone named **musicmania.in**.

- **Configuration Steps (BIND on Linux)**

Step 1: Install BIND

```
sudo apt update
sudo apt install bind9 bind9utils bind9-doc
```

Step 2: Create a Zone Entry

```
sudo nano /etc/bind/named.conf.local
```

Add:

```
zone "musicmania.in" {
    type master;
    file "/etc/bind/db.musicmania.in";
};
```

Step 3: Create the Zone File

```
sudo cp /etc/bind/db.local /etc/bind/db.musicmania.in
```

Edit:

```
sudo nano /etc/bind/db.musicmania.in
```

Example configuration:

```
$TTL 604800
@ IN SOA ns.musicmania.in. admin.musicmania.in. (
    2
    604800
    86400
    2419200
    604800 )
```

```
@    IN NS   ns.musicmania.in.  
ns   IN A    192.168.1.50  
www  IN A    192.168.1.50
```

Step 4: Restart DNS Service

sudo systemctl restart bind9

```
student@ubuntu:~$ sudo systemctl restart bind9  
student@ubuntu:~$ sudo systemctl status bind9  
● bind9.service - BIND Domain Name Server  
   Loaded: loaded (/lib/systemd/system/bind9.service; enabled; vendor preset: enabled)  
   Active: active (running) since Wed 29 May 2024 10:45:12 AM IST; 3s ago  
     Docs: man:named(8)  
    Main PID: 1152 (named)  
      Tasks: 6 (limit: 2271)  
    Memory: 12.6M  
       CPU: 58ms  
    CGroup: /system.slice/bind9.service  
            └─1152 /usr/sbin/named -f -u bind  
  
May 29 10:45:12 ubuntu systemd[1]: Starting BIND Domain Name Server...  
May 29 10:45:12 ubuntu systemd[1]: Started BIND Domain Name Server.  
student@ubuntu:~$
```

2. Add an A record in your DNS server configuration to map `www.musicmania.in` to the IP address `192.168.1.50`, then use the `nslookup` or `dig` command to verify the mapping works.

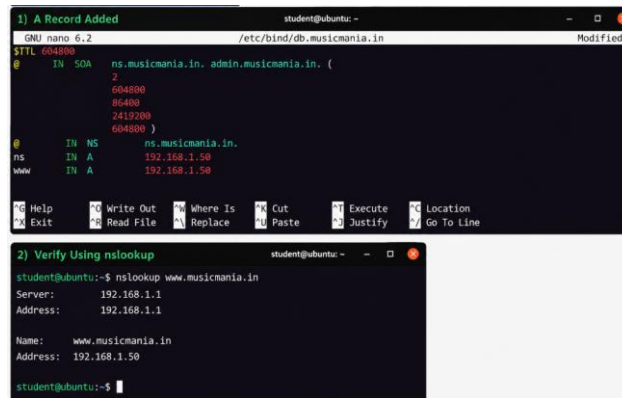
ANS :

- **A Record**
www IN A 192.168.1.50
- **Verify Using nslookup**
nslookup www.musicmania.in
- **Sample Output**
Server: 192.168.1.1
Address: 192.168.1.1

Name: www.musicmania.in
Address: 192.168.1.50

- **Explanation**

The A record maps the hostname www.musicmania.in to the IP address **192.168.1.50**. When users enter the domain name, the DNS server returns this IP address.



```
1) A Record Added
student@ubuntu: ~
GNU nano 6.2 /etc/bind/db.musicmania.in Modified
$TTL 604800
@ IN SOA ns.musicmania.in. admin.musicmania.in. (
    2
    604800
    86400
    2419200
    604800 )
@ IN NS ns.musicmania.in.
ns IN A 192.168.1.50
www IN A 192.168.1.50

2) Verify Using nslookup
student@ubuntu:~$ nslookup www.musicmania.in
Server: 192.168.1.1
Address: 192.168.1.1

Name: www.musicmania.in
Address: 192.168.1.50
student@ubuntu:~$
```

3. Simulate a typo by adding a CNAME record for play.musicmania.in that points to www.musicmania.in, then test if accessing play.musicmania.in resolves to the correct IP address.

ANS :

➤ **Add a CNAME Record**

- **Configuration**

Add the following line to the zone file:
play IN CNAME www.musicmania.in.

- **Restart DNS**

sudo systemctl restart bind9

- **Verify**

nslookup play.musicmania.in

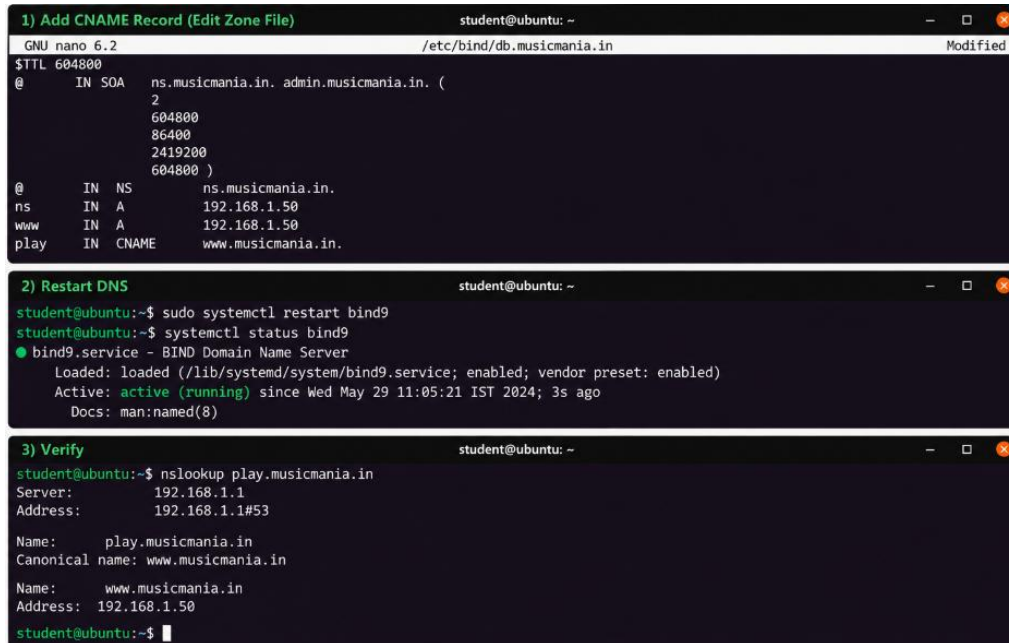
- **Sample Output**

Name: play.musicmania.in
Canonical name: www.musicmania.in

Name: www.musicmania.in
Address: 192.168.1.50

- **Explanation**

A CNAME (Canonical Name) record creates an alias. Here, **play.musicmania.in** points to www.musicmania.in, which then resolves to **192.168.1.50**.



```
1) Add CNAME Record (Edit Zone File)
student@ubuntu: ~
GNU nano 6.2 /etc/bind/db.musicmania.in Modified
$TTL 604800
@      IN  SOA  ns.musicmania.in. admin.musicmania.in. (
        2
        604800
        86400
        2419200
        604800 )
@      IN  NS   ns.musicmania.in.
ns     IN  A    192.168.1.50
www    IN  A    192.168.1.50
play   IN  CNAME www.musicmania.in.

2) Restart DNS
student@ubuntu: ~
student@ubuntu:~$ sudo systemctl restart bind9
student@ubuntu:~$ systemctl status bind9
● bind9.service - BIND Domain Name Server
   Loaded: loaded (/lib/systemd/system/bind9.service; enabled; vendor preset: enabled)
   Active: active (running) since Wed May 29 11:05:21 IST 2024; 3s ago
     Docs: man:named(8)

3) Verify
student@ubuntu:~$ nslookup play.musicmania.in
Server:      192.168.1.1
Address:     192.168.1.1#53

Name:        play.musicmania.in
Canonical name: www.musicmania.in
Name:        www.musicmania.in
Address:     192.168.1.50
student@ubuntu:~$
```

4. Your DNS server is not resolving the domain musicmania.in as expected. Document the step-by-step troubleshooting process you would follow, including checking configuration files, restarting the DNS service, and testing with nslookup or dig. Hint : Include at least three different troubleshooting commands or checks in your documentation.

ANS :

Problem

The domain **musicmania.in** is not resolving correctly.

Step 1: Check DNS Configuration

- Verify that the zone configuration is correct.
- Command:
 - `sudo named-checkconf`
- Expected Result:
- No errors should be displayed.

Step 2: Verify Zone File

- Check for syntax errors in the zone file.
- Command:
 - `sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in`
- Expected Result : OK

Step 3: Restart the DNS Service

- After making changes, restart BIND.
- Command:
- `sudo systemctl restart bind9`
- Check service status:
- `sudo systemctl status bind9`
- Expected Result:
- The service should show **active (running)**.

Step 4: Test Domain Resolution

- Use **nslookup**.
- `nslookup www.musicmania.in`
- or use **dig**.
- `dig www.musicmania.in`
- Expected Result:
- `www.musicmania.in`
Address: 192.168.1.50

Step 5: Check Log Files

- If the problem continues, inspect DNS logs.
- `sudo journalctl -u bind9`
- or
- `sudo tail -f /var/log/syslog`
- Look for errors such as:
- Zone file syntax error
- Permission denied
- Service failed to start

Step 6: Verify Firewall (Optional)

- Ensure DNS traffic is allowed.
- `sudo ufw status`
- If necessary:
- `sudo ufw allow 53`

1) Check DNS Configuration

```
student@ubuntu:~$ sudo named-checkconf
```

```
student@ubuntu:~$
```

Expected Result: No errors should be displayed.

2) Verify Zone File

```
student@ubuntu:~$ sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in
```

```
zone musicmania.in/IN: loaded serial 2
```

```
OK
```

```
student@ubuntu:~$
```

Expected Result: OK

3) Restart the DNS Service & Check Status

```
student@ubuntu:~$ sudo systemctl restart bind9
```

```
student@ubuntu:~$ sudo systemctl status bind9
```

```
● bind9.service - BIND Domain Name Server
```

```
Loaded: loaded (/lib/systemd/system/bind9.service; enabled; vendor preset: enabled)
```

```
Active: active (running) since Wed May 29 11:15:45 IST 2024; 2s ago
```

```
Docs: man:named(8)
```

```
Main PID: 1234 (named)
```

```
Tasks: 6 (limit: 2277)
```

```
Memory: 12.1M
```

```
CGroup: /system.slice/bind9.service
```

```
└─1234 /usr/sbin/named -f -u bind
```

```
student@ubuntu:~$
```

Expected Result: The service should show active (running).

4) Test Domain Resolution (nslookup & dig)

```
student@ubuntu:~$ nslookup www.musicmania.in
```

```
Server: 192.168.1.1
```

```
Address: 192.168.1.1#53
```

```
Name: www.musicmania.in
```

```
Address: 192.168.1.50
```

```
student@ubuntu:~$
```

```
student@ubuntu:~$ dig www.musicmania.in
```

```
; <<>> DiG 9.18.1-ubuntu1.1-Ubuntu <<>> www.musicmania.in
```

```
;; global options: +cmd
```

```
;; Got answer:
```

```
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 12345
```

```
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 1, ADDITIONAL: 1
```

```
;; QUESTION SECTION:
```

```
;www.musicmania.in. IN A
```

```
;; ANSWER SECTION:
```

```
www.musicmania.in. 604800 IN A 192.168.1.50
```

```
;; SERVER: 192.168.1.1#53(192.168.1.1)
```

```
;; WHEN: Wed May 29 11:16:30 IST 2024
```

```
;; MSG SIZE rcvd: 81
```

```
student@ubuntu:~$
```

Expected Result: www.musicmania.in
Address: 192.168.1.50

5) Check Log Files

```
student@ubuntu:~$ sudo journalctl -u bind9 -n 10
```

```
May 29 11:15:45 ubuntu named[1234]: zone musicmania.in/IN: loaded serial 2
```

```
May 29 11:15:45 ubuntu named[1234]: all zones loaded
```

```
May 29 11:15:45 ubuntu named[1234]: running
```

```
May 29 11:15:45 ubuntu systemd[1]: Started BIND Domain Name Server.
```

```
student@ubuntu:~$
```

```
student@ubuntu:~$ sudo tail -f /var/log/syslog
```

```
May 29 11:16:02 ubuntu named[1234]: client 192.168.1.100#54321: query:
```

```
www.musicmania.in IN A +E (192.168.1.1)
```

```
May 29 11:16:02 ubuntu named[1234]: client 192.168.1.100#54321: query:
```

```
www.musicmania.in IN A +E (192.168.1.1)
```

```
student@ubuntu:~$
```

Look for errors such as:

- Zone file syntax error
- Permission denied
- Service failed to start

6) Verify Firewall (Optional)

```
student@ubuntu:~$ sudo ufw status
```

```
Status: active
```

To	Action	From
--	----	----
53/tcp	ALLOW	Anywhere
53/udp	ALLOW	Anywhere
22/tcp	ALLOW	Anywhere
22/tcp (v6)	ALLOW	Anywhere (v6)
53 (v6)	ALLOW	Anywhere (v6)
53/udp (v6)	ALLOW	Anywhere (v6)
22/tcp (v6)	ALLOW	Anywhere (v6)

```
student@ubuntu:~$
```

OR

```
student@ubuntu:~$ sudo ufw allow 53
```

```
Rules updated
```

```
Rules updated (v6)
```

```
student@ubuntu:~$ sudo ufw status
```

```
Status: active
```

To	Action	From
--	----	----
53/tcp	ALLOW	Anywhere
53/udp	ALLOW	Anywhere
22/tcp	ALLOW	Anywhere

```
student@ubuntu:~$
```